Category: Internet Measurement



A3-PS010-TC514

Evidence-Based IP Network-Type Classification

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The Problem: IP Classification in the Age of Churn

Why can an IP address be difficult to classify reliably?

- **Static or coarse databases**

  Commercial IP databases can rely on static or coarse-grained classifications.
- **Tor and VPN endpoints change**

  Tor exit relays and VPN endpoints can change over time.
- **BGP routing changes**

  BGP routing changes can alter the origin Autonomous System (AS) associated with a prefix.
- **Shared addressing (CGNAT)**

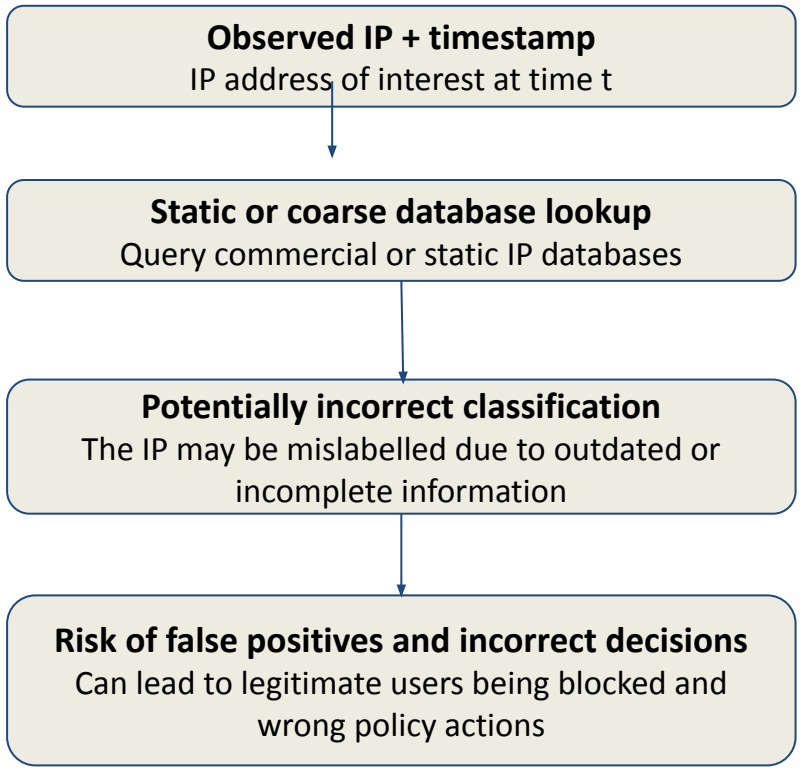
  users can share one public IP. Carrier-grade NAT (CGNAT) and shared addressing mean multiple
- **Hosting is not proof of VPN usage**

  Hosting infrastructure does not automatically mean that an IP is a VPN or proxy endpoint.

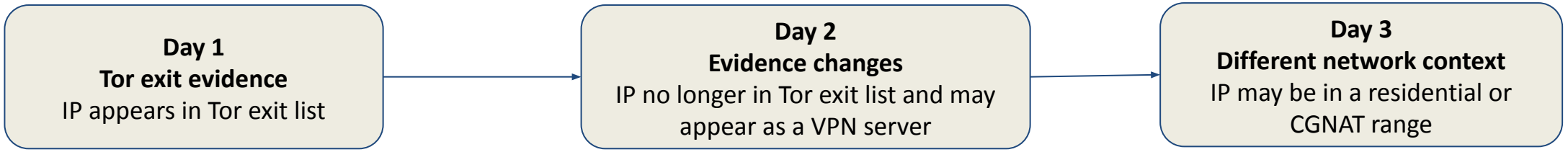
Proposed Objective

Develop an evidence-based classifier that evaluates an IP address against evidence valid at the observation timestamp and returns an auditable classification.

Why static lookups can be misleading



IP evidence can change over time



Today's evidence cannot automatically establish yesterday's classification.

Background: Why Timestamped Evidence Matters

An IP address can belong to different network contexts over time.

Tor relay changes

- Tor exit lists and Onionoo provide first-seen and last-seen timestamps.
- Tor exit relays can rotate over time.
- An IP may be a Tor exit at one time and not at another.

BGP and ASN changes

- The origin Autonomous System (AS) for a prefix can change over time.
- RouteViews and RIPE RIS provide historical routing data.
- Current ASN information may not reflect past observations.

Hosting is not proof of VPN usage

- Cloud providers, universities and CDNs use hosting ASNs.
- Most hosting IPs are not VPN or proxy endpoints.
- ASN information alone cannot determine the endpoint type.

Shared addressing and CGNAT

- Carrier-grade NAT (CGNAT) allows multiple users to share one public IP address (RFC 6269).
- We cannot claim a direct connection to a person or device for an unlabelled IP.
- The classifier labels networks, not individuals.

Conservative classification principle

- **Sufficient evidence**
  - A valid, time-relevant match from a trusted source supports a positive classification (Known Tor exit or Known VPN/proxy endpoint).
- **Insufficient, stale or conflicting evidence**
  - If the evidence is missing, stale, ambiguous or conflicting, the output is Unknown with a reason code.

Literature & RFC Review

Only the most relevant standards, data sources and studies.

Source or Standard	Relevance to PS-010	Limitation Addressed
RFC 6269	IP address sharing	An IP does not uniquely identify a user
RFC 4271	BGP routing	Origin-AS information must be interpreted in its routing context
RFC 6598	Shared IPv4 address space	Shared addressing complicates attribution
Tor Project exit list and Onionoo	Tor exit evidence	Relay status must be time-relevant
RouteViews and RIPE RIS	Historical BGP information	Current routing may not represent past observations
Mi et al. (IEEE S&P 2019)	Residential IP proxy services	IP and ASN context alone may not establish endpoint type
Winter et al. (PETS 2014)	Tor exit relay behaviour	Relay characteristics and behaviour need careful interpretation

Research Gap & Contribution

What PS-010 Improves

- Time-based, multi-source evidence with traceable provenance.
- Conservative classification with an **Unknown** option.
- Reproducible evidence snapshots using SHA-256 hashes.
- Per-class evaluation and false-positive testing on ordinary hosting.

Key Contribution: An auditable, reproducible rule-based pipeline that classifies IP observations without treating weak evidence as proof.

Proposed Solution

A conservative, auditable rule pipeline over timestamped evidence.

Pipeline

- Input: (IP, timestamp).
- Evidence collectors: Tor exit list, BGP/ASN origin as-of t, hosting ranges, lab VPN/proxy endpoints.
- Each collector returns a record: source, snapshot date, SHA-256, match.
- Rule engine resolves the records by priority and conflict rules.
- Output: label + evidence JSON + reason code.
- Evaluator: per-class precision, recall, coverage.

Decision rules (priority order)

Rule	Condition	Output
R0	Invalid or excluded address (according to input policy)	Unknown
R1	Valid Tor exit evidence applicable at time t	Known Tor exit
R2	Valid match in known VPN/proxy endpoint list applicable at time t	Known VPN/Proxy endpoints
R3	Evidence is stale, ambiguous or conflicting	Unknown
R4	Hosting ASN alone or no sufficient positive evidence	Unknown

Important principle

A hosting ASN alone must never establish that an IP is a VPN or proxy endpoint.

Optimization Proposed

Before vs after: what changes compared with a static IP-reputation table.

Aspect	Before (static database)	After (proposed)
Time handling	Today's list applied to old observations.	As-of-time lookup in pinned, dated snapshots.
Hosting ASN	Hosting treated as VPN, causing false positives.	Hosting is evidence only and never labels an IP on its own.
Uncertainty	Forced guess, often 'residential'.	Explicit Unknown with a machine-readable reason code.
Output	Label only.	Label + evidence + provenance JSON per decision.
Reproducibility	Opaque vendor updates.	SHA-256 pinned snapshots; reruns give identical results.
Evaluation	Accuracy only.	Per-class precision/recall, coverage, hosting-negative FPR, ablations.

- Extra analysis: precision-vs-coverage curve as rules are tightened or loosened, plus ablation without Tor, ASN or lab data.

Expected benefits

- More transparent classification decisions.
- Better reproducibility across repeated evaluations.
- Explicit handling of uncertainty and conflicting evidence.
- Direct testing of the risk of misclassifying ordinary hosting infrastructure.
- Easier investigation of why a particular label was assigned..

Solution Architecture & Design

Component view of the classification pipeline.

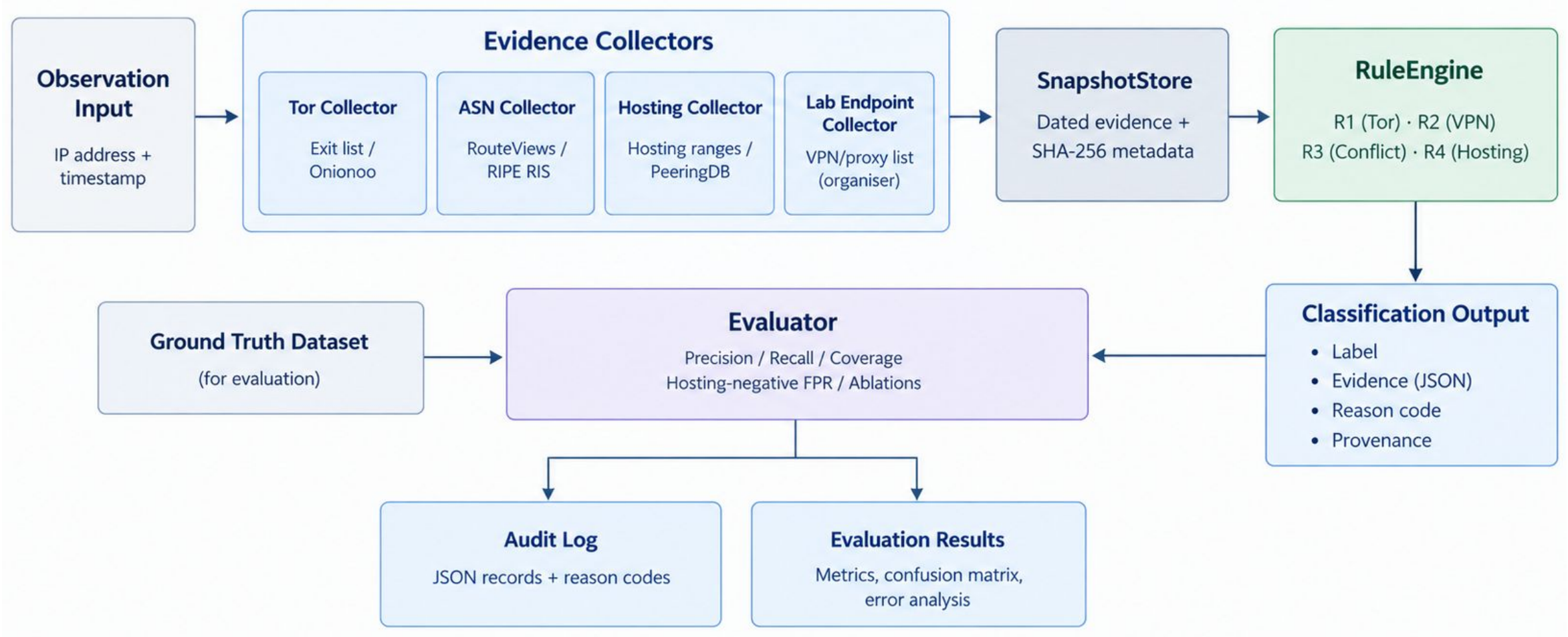


Figure: System Architecture UML Diagram

Data Flow Diagram

Level-1 DFD: how an observation becomes a label with evidence.

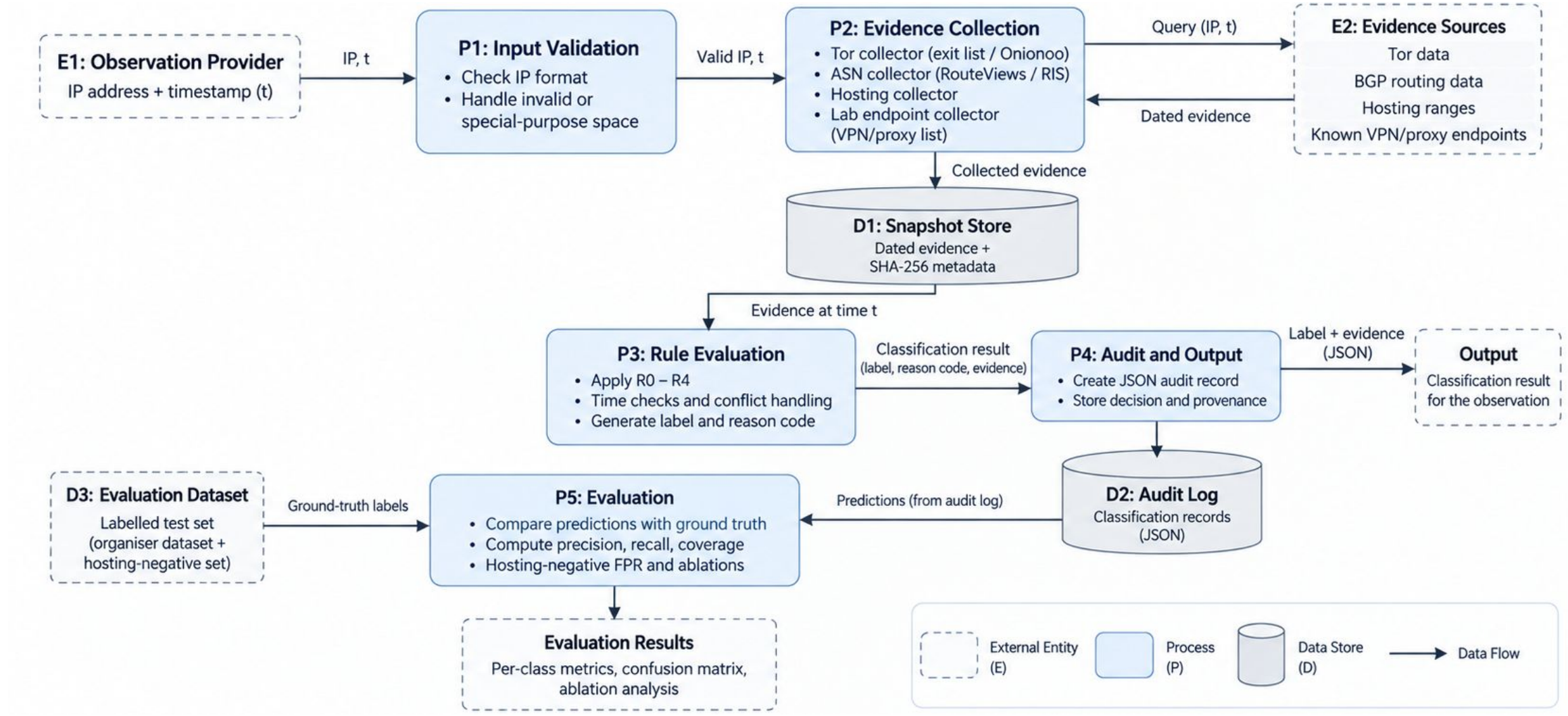


Figure: Data Flow Diagram (DFD)

Implementation Roadmap & Deliverables

Milestones and methodology: build, evaluate, document, demonstrate.

Milestone	Planned activities	Deliverable
Setup	Repository, evidence sources and snapshot storage	Project foundation
Development	Evidence collectors, rule engine and audit logging	Classification pipeline
Evaluation	Testing, precision/recall, coverage and false-positive analysis	Evaluation report
Finalization	Documentation, working demo and required submissions	Submission-ready repository
Final	Upload this PDF, transfer repo ownership to aiori-hackathon, live demo with pseudocode snippet.	Report, demo, certificate-ready

Bibliography

References

1. Ford, M. et al. RFC 6269: Issues with IP Address Sharing. Internet Engineering Task Force (IETF), 2011. https://www.rfc-editor.org/rfc/rfc6269

2. Rekhter, Y., Li, T., and Hares, S. (eds.). RFC 4271: A Border Gateway Protocol 4 (BGP-4). IETF, 2006. https://www.rfc-editor.org/rfc/rfc4271

3. Weil, J. et al. RFC 6598: IANA-Reserved IPv4 Prefix for Shared Address Space. IETF, 2012. https://www.rfc-editor.org/rfc/rfc6598

4. The Tor Project. Tor Bulk Exit List and Onionoo Protocol Documentation. https://metrics.torproject.org/

5. University of Oregon. RouteViews Project. https://www.routeviews.org/

6. RIPE NCC. Routing Information Service (RIS). https://www.ripe.net/analyse/internet-measurements/routing-information-service-ris/

7. Mi, X. et al. Resident Evil: Understanding Residential IP Proxy as a Dark Service. IEEE Symposium on Security and Privacy, 2019.

8. Winter, P. et al. Spoiled Onions: Exposing Malicious Tor Exit Relays. Proceedings of Privacy Enhancing Technologies, 2014.

Optional additional sources

Include these only if you actually use them in the project:

9. IETF. RFC 6890: Special-Purpose IP Address Registries. https://www.rfc-editor.org/rfc/rfc6890

10. CAIDA. BGPStream and AS Rank resources. https://www.caida.org/

Reference-checking note: Before submission, verify the exact publication metadata for references 7 and 8 and ensure each source supports the specific claim cited in the presentation. RFC 6890 concerns special-purpose IP address registries; it should not be cited as evidence for VPN detection.

Thank You for Your Attention!

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